3.6

ADULTS

Febrile Neutropenia

Haematopoietic Stem Cell Transplantation (HSCT), Haematology and Oncology	
Febrile neutropenia	Neutrophils < 0.5 x 10^9/L, or < 1 x 10^9/L with a predicted decline to < 0.5 x 10^9/L + fever ≥ 38ºC.
Low risk patient	Anticipate < 7 days profound neutropenia, no comorbidities, can take orally, accessible to hospital care (MASCC score ≥ 21).
High risk patient	Anticipate ≥ 7 days profound neutropenia (ANC < 1 x 109/L) ± significant medical comorbid conditions (MASCC score < 21).

CHAPTER 3.6	FEBRILE NEUTROPENIA		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Low risk (Outpatient)			
	Co-amoxiclav 625 mg PO q8H ± Ciprofloxacin 750 mg PO	Clindamycin 900 mg PO q8H ± Ciprofloxacin 750 mg PO	• First dose to be administered in the clinic or Emergency Department. • Duration: Afebrile for at least 48 hours or clinically stable and ANC count > 0.5 x 10^9/L in the absence of site-specific

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Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	q12H*	q12H*	infection • *If colonised by *Pseudomonas aeruginosa* previously (within the last 6 months).
High risk – Stable patient			
Initial fever	Cefepime 2 g IV q8H ± Amikacin 15 mg/kg IV q24H ± Vancomycin 15 mg/kg (max 1 g) q8-12H*	Piperacillin/Tazobactam 4.5 g IV q6H ± Amikacin 15 mg/kg IV q24H **OR** Vancomycin 15 mg/kg (max 1 g) q8-12H*	• Duration: Afebrile for at least 48 hours or clinically stable and ANC count > 0.5 x 10^9/L. • *Vancomycin is not recommended as a standard part of the initial antibiotic regimen. • Only consider for: known MRSA coloniser, suspected catheter-related infection, skin, or soft-tissue infection, or in shock. • Stop after 48 hours if there is no evidence of gram-positive cocci.
Persistent fever or new fever after 4-7 days or deterioration of clinical signs Consider VRE, resistant	Meropenem 1 g IV q8H ± Vancomycin 15 mg/kg (max 1 g) q8-12H*		• Consider imaging studies and further microbiologic testing. • Please refer to ID consult.

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Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
gram-negative bacilli. Consider invasive fungal infections if new suggestive clinical signs or symptoms and/or if fever persists ≥ 4-7 days. High risk patients (neutropenia >10 days, allogeneic HCT recipients, high-dose corticosteroids). Received none or only Fluconazole antifungal prophylaxis. Sinus and/or chest CT not suggestive of fungal infection.	**Add antifungal to antibiotic above:** Micafungin 100 mg IV q24H **OR** Voriconazole 6 mg/kg IV q12H on Day 1, then 4 mg/kg IV q12H **(if mould is suspected)**	**Add antifungal to antibiotic above:** Amphotericin B Deoxycholate 0.7 - 1 mg/kg IV q24H **OR** Amphotericin B Lipid Complex 5 mg/kg IV q24H	• Repeat fungal culture EOD if positive for candidemia and no evidence of invasive disease. • Duration: • Yeast (e.g., Candida): at least 2 weeks after first negative blood culture. • Mould (e.g., Aspergillus): minimum of 12 weeks. • If there is no positive culture but is CT scan / biopsy proven: continue until off immunosuppressant or no further chemotherapy.
CT suggestive of fungal infection.	**Add antifungal to antibiotic above:** Amphotericin B Deoxycholate 0.7-1 mg/kg IV q24H	**Add antifungal to antibiotic above:** Posaconazole M/R 300 mg PO q12H for 2 days then 300 mg q24H	• *Isavuconazole is not available in UMMC.

CHAPTER 3.6	FEBRILE NEUTROPENIA		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	(if sinus involvement) **OR** Voriconazole 6 mg/kg IV q12H on Day 1, then 4 mg/kg IV q12H **(if NO sinus involvement)**	**OR** Isavuconazole* 200 mg q8H on Day 1 then 200 mg q24H	
Received Voriconazole or Posaconazole prophylaxis	**Add antifungal to antibiotic above:** Amphotericin B Deoxycholate 0.7-1 mg/kg IV q24H	**Add antifungal to antibiotic above:** Isavuconazole* 200 mg q8H on Day 1 then 200 mg q24H	
High risk – Unstable patient			
Initial fever	Meropenem 1 g IV q8H ± Vancomycin 15 mg/kg (max 1 g) q8-12H	Cefepime 2 g IV q8H ± Amikacin 15 mg/kg IV q24H ± Vancomycin 15 mg/kg (max 1 g) q8-12H	• Duration of therapy depends on clinical course, neutropenia recovery, toxicity, and ID consult.

CHAPTER 3.6	FEBRILE NEUTROPENIA		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Persistent fever or new fever after 4 - 7 days	**Add antifungal as above** (for stable patients)		

Antimicrobial Prophylaxis in Patients with Haematological Malignancies

Please refer to **Chapter 3.18: Non-surgical Prophylaxis**.